

THEORETICAL QUESTIONS



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Question 1: Backpropagation in Feedforward Neural Networks

Model Definition

- **Input:** $x \in \mathbb{R}^d$
- **Hidden layer:**

$$z_1 = W_1 x + b_1, \quad h = \sigma(z_1) = \text{ReLU}(z_1)$$

where $W_1 \in \mathbb{R}^{m \times d}$, $b_1 \in \mathbb{R}^m$

- **Output layer:**

$$\hat{y} = w_2^\top h + b_2$$

where $w_2 \in \mathbb{R}^m$, $b_2 \in \mathbb{R}$

- **Loss function (Squared Error):**

$$L = \frac{1}{2}(y - \hat{y})^2$$

1. Forward Pass

The full composite function from input x to prediction $\hat{y}$ is

$$\hat{y} = w_2^\top \text{ReLU}(W_1 x + b_1) + b_2$$

2. Backward Pass (Output Layer)

First compute the derivative of the loss with respect to the prediction:

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

Since

$$\hat{y} = w_2^\top h + b_2$$

we have

$$\frac{\partial \hat{y}}{\partial w_2} = h$$

Applying the chain rule:

$$\boxed{\frac{\partial L}{\partial w_2} = (\hat{y} - y) h}$$

(For completeness)

$$\frac{\partial L}{\partial b_2} = \hat{y} - y$$

3. Backward Pass (Hidden Layer)

Gradient w.r.t. hidden activations

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h} = (\hat{y} - y) w_2$$

Through ReLU activation

For ReLU,

$$\sigma'(z_1)_i = \mathbf{1}(z_{1,i} > 0)$$

Thus

$$\frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial h} \odot \sigma'(z_1) = (\hat{y} - y) (w_2 \odot \sigma'(z_1))$$

Define the hidden-layer error term:

$$\delta_1 = (\hat{y} - y) (w_2 \odot \sigma'(z_1))$$

Gradient w.r.t. input weights W_1

Since

$$z_1 = W_1 x + b_1$$

the gradient is an outer product:

$$\boxed{\frac{\partial L}{\partial W_1} = \delta_1 x^\top}$$

(For completeness)

$$\frac{\partial L}{\partial b_1} = \delta_1$$

Summary of Gradients

$$\frac{\partial L}{\partial w_2} = (\hat{y} - y) h$$

$$\frac{\partial L}{\partial W_1} = (\hat{y} - y) (w_2 \odot \sigma'(W_1 x + b_1)) x^\top$$

This is the full **backpropagation derivation** for a one-hidden-layer FFNN with ReLU and squared error loss.

✓ Question 2: The Mathematics of Self-Attention

Given (2 tokens, $d_k = 4$):

$$Q = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Attention scores

First compute $K^\top$:

$$K^\top = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Now:

$$S = QK^\top = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}$$

Scaled softmax

Scale by $\sqrt{d_k} = \sqrt{4} = 2$

$$\tilde{S} = \frac{S}{\sqrt{d_k}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 \\ 1 & 0 \end{pmatrix}$$

Apply softmax row-wise

Row 1:

$$\text{softmax}([0.5, 0.5]) = [0.5, 0.5]$$

Row 2:

$$\text{softmax}([1, 0]) = \left[\frac{e^1}{e^1 + e^0}, \frac{e^0}{e^1 + e^0} \right] = \left[\frac{e}{e + 1}, \frac{1}{e + 1} \right] \approx [0.7311, 0.2689]$$

So the attention weight matrix is

$$A = \begin{pmatrix} 0.5 & 0.5 \\ \frac{e}{e+1} & \frac{1}{e+1} \end{pmatrix} \approx \begin{pmatrix} 0.5 & 0.5 \\ 0.7311 & 0.2689 \end{pmatrix}$$

Context Vector

Compute:

$$Z = \begin{pmatrix} 0.5 & 0.5 \\ \frac{e}{e+1} & \frac{1}{e+1} \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

First row of Z

$$Z_1 = 0.5 V_1 + 0.5 V_2 = 0.5[1, 2] + 0.5[3, 4] = [2, 3]$$

Second row of Z (exact + approx)

Let $a = \frac{e}{e+1}$ and $b = \frac{1}{e+1}$.

$$Z_2 = a V_1 + b V_2 = a[1, 2] + b[3, 4]$$

Component-wise:

$$Z_{2,1} = \frac{e + 3}{e + 1} \approx 1.5379, \qquad Z_{2,2} = \frac{2e + 4}{e + 1} \approx 2.5379$$

So

$$Z = \begin{pmatrix} 2 & 3 \\ \frac{e+3}{e+1} & \frac{2e+4}{e+1} \end{pmatrix} \approx \begin{pmatrix} 2 & 3 \\ 1.5379 & 2.5379 \end{pmatrix}$$

Interpretation (How much does token 1 attend to token 2?)

The amount of attention token 1 gives to token 2 is the weight:

$$A_{12} = 0.5$$

So **the first token takes 50% of its information from the second token’s value vector** $V_2 = [3, 4]$ (and 50% from its own $V_1 = [1, 2]$), producing:

$$Z_1 = 0.5V_1 + 0.5V_2 = [2, 3].$$